# Extra Practice 1: Money Sense: Rounding, Estimating and Using Your Calculator

## A. Skills

**1.** Name the place of the 7 in each amount, and write its value in dollars.
(a) \$4.72 (b) \$0.07 (c) \$1.007

**2.** Round each amount.

| Amount | Nearest dollar | Nearest ten cents | Nearest cent |
|---|---|---|---|
| \$5.638 | | | |
| \$19.451 | | | |
| \$2.294 | | | |
| \$99.996 | | | |

**3.** Work out each by hand, using BEDMAS.
(a) $12 - 2 \times 5$ (b) $(12 - 2) \times 5$ (c) $18 \div (4 + 2)$ (d) $18 \div 4 + 2$

**4.** Round each calculator display to the nearest cent and write it as money.
(a) `7.1449` (b) `15.66666667` (c) `2.005` (d) `39.9951`

_____ _____ _____ _____

**5.** Find the change.
(a) A \$6.85 sandwich, paid with a \$10 bill
(b) A \$63.25 grocery order, paid with a \$100 bill

## B. Applying

**6.** At the hardware store, Lena buys items for \$12.89, \$4.35, \$7.69 and \$2.19.
(a) Find a front-end estimate of the total.

(b) Find the exact total.

**7.** For back-to-school, Owen picks out jeans for \$39.99, a shirt for \$18.49, shoes for \$64.95 and socks for \$9.79. He has \$150.
(a) Use a rounding estimate to decide if he has enough money.

(b) Find the exact total.

**8.** At a café, Rosa orders 2 sandwiches at \$7.45 each, 3 cookies at \$1.65 each and a large juice for \$3.95.

(a) Write one calculation for the total, and find it.

(b) Check your answer with an estimate.

**9.** Zoe babysits for 4.5 hours at \$14.50 per hour. Her friend's calculator says she earned \$652.50.

(a) Estimate Zoe's pay. Is \$652.50 reasonable?

(b) Find Zoe's exact pay.

**10.** Raj fills his truck with 48.6 L of gas at \$1.429 per litre. Estimate the cost, then find the exact cost to the nearest cent.

**11.** Tessa buys 7 bags of dog treats at \$5.89 each. The best estimate of the total is

**A.** \$35 **B.** \$42 **C.** \$70 **D.** \$4.20

**12. Numerical Response** A 4-pack of AA batteries costs \$7.98. The cost of one battery, to the nearest cent, is \$_____.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**13.** Your total at the drugstore is \$27.62.

(a) What do you pay if you use cash? _____

(b) What is your change from two \$20 bills?

(c) What do you pay if you use debit? _____

**14.** Six roommates share a \$100.00 internet bill equally.

(a) What does the calculator show for $100 \div 6$? _____

(b) If every roommate pays \$16.67, how much is collected?

(c) Show a fair way to collect exactly \$100.00.

**15.** A receipt has a total of \$24.30 for 2 phone chargers at \$5.85 each and one phone case. The price of the case is smudged. Write one calculation for the price of the case, and find it.

**16.** Maya earns \$16.25 per hour and is paid only for whole hours. How many hours must she work to earn at least \$400? Explain why you round the way you do.

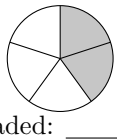
## Extra Practice 2: Fractions Refresher: Equivalent Fractions and Simplest Form

### A. Skills

**1.** Write the fraction of each picture that is shaded, and the fraction that is not shaded.

(a) 

Shaded: _____ Not shaded: _____

(b) 

Shaded: _____ Not shaded: _____

**2.** Find the missing number.

(a) $\frac{4}{7} = \frac{\square}{21}$ (b) $\frac{6}{9} = \frac{2}{\square}$ (c) $\frac{5}{12} = \frac{25}{\square}$ (d) $\frac{16}{20} = \frac{\square}{5}$

**3.** Find the greatest common factor (GCF) of each pair of numbers.

(a) 15 and 25 (b) 24 and 32 (c) 18 and 42

**4.** Write each fraction in simplest form.

(a) $\frac{10}{12}$ (b) $\frac{15}{40}$ (c) $\frac{27}{36}$ (d) $\frac{32}{48}$ (e) $\frac{60}{75}$ (f) $\frac{44}{100}$

**5.** Change each fraction to a decimal (round to the nearest thousandth if it repeats), and each decimal to a fraction in simplest form.

(a) $\frac{1}{8}$ (b) $\frac{9}{20}$ (c) $\frac{5}{6}$ (d) 0.65 (e) 0.25

**6.** Write $<$, $>$ or $=$ in each box. Show your method.

(a) $\frac{5}{6} \square \frac{7}{9}$ (b) $\frac{3}{8} \square \frac{2}{5}$ (c) $\frac{10}{16} \square \frac{5}{8}$

**7.** Find each amount.

(a) $\frac{3}{4}$ of 60 (b) $\frac{2}{9}$ of 54 (c) $\frac{4}{5}$ of \$35

### B. Applying

**8.** A community theatre has 240 seats. For Friday's show, 180 tickets were sold.

(a) What fraction of the seats were sold? Write it in simplest form.

(b) What fraction of the seats were empty?

**9.** Three high school soccer teams finished the season with these records: the Hawks won 12 of 20 games, the Bears won 14 of 25 games, and the Lynx won 9 of 15 games. Which team had the best record? Explain.

**10.** A muffin recipe needs $\frac{3}{4}$ cup of sugar. Liam has $\frac{2}{3}$ cup of sugar left. Does he have enough? Show how you know.

**11.** Hana's phone plan has 12 GB of data a month. She has used $\frac{5}{8}$ of it.

(a) How many gigabytes has she used?

(b) How many gigabytes does she have left?

**12.** A pair of runners costs \$45. Mateo has saved $\frac{3}{5}$ of the price. How much has he saved?

**A.** \$27.00 **B.** \$75.00 **C.** \$15.00 **D.** \$9.00

**13. Numerical Response** Sienna got 18 out of 24 on a quiz. Her mark is equivalent to $\frac{\square}{100}$. What number goes in the box?

| | | | |
|---|---|---|---|
| | | | |

### C. Stretch

**14.** Marcus gets a paycheque of \$350. He spends $\frac{2}{5}$ of it on groceries. Then he saves $\frac{1}{3}$ of what is left.

(a) How much does he save?

(b) What fraction of the whole paycheque does he save? Write it in simplest form.

**15.** Find a fraction that is between $\frac{3}{5}$ and $\frac{2}{3}$. Show how you know it is between them.

**16.** Without a calculator, decide which is bigger, $\frac{7}{8}$ or $\frac{8}{9}$. (Hint: how much is each one missing to make 1 whole?)

## Extra Practice 3: Ratios

### A. Skills

**1.** A bouquet has 6 roses, 4 lilies and 2 daisies. Write each ratio three ways (colon, words, fraction), then in simplest form.
(a) roses to lilies
(b) daisies to all flowers

**2.** Write each ratio in simplest form.
(a) 15 : 25 (b) 42 : 6 (c) 27 : 36 (d) 50 : 125

**3.** Write each ratio in simplest form, using whole numbers in the same unit.
(a) 0.5 : 2 (b) 3.2 : 4.8 (c) 90 s : 3 min (d) 1.2 m : 40 cm (e) 400 mL : 1.4 L

**4.** Complete the ratio table for the ratio 5 : 2.

| First term | 5 | 10 | | 25 | |
|---|---|---|---|---|---|
| Second term | 2 | | 8 | | 3 |

**5.** Are the two ratios equivalent? Write yes or no.
(a) 3 : 4 and 9 : 12 (b) 2 : 5 and 5 : 2 (c) 6 : 8 and 15 : 20 (d) 7 : 3 and 21 : 6

### B. Applying

**6.** Kofi works 4 h 30 min at a movie theatre on Friday and 6 h on Saturday. Write the ratio of his Friday time to his Saturday time in simplest form.

**7.** Kayla made 18 of her 30 free throws this season.
(a) Write the ratio of shots made to shots missed in simplest form.
(b) Write the ratio of shots made to shots taken in simplest form.

**8.** A cleaning spray is mixed 1 part vinegar to 4 parts water.

(a) How much water goes with 750 mL of vinegar? Give your answer in litres.

(b) How much vinegar is in a full 1 L spray bottle?

**9.** Two sisters share $350 from a garage sale in the ratio 3 : 4. How much does each get?

**10.** A 1.2 kg bag of snack mix has pretzels, cereal and nuts in the ratio 3 : 4 : 1 by mass. How many grams of each are in the bag?

**11.** Which ratio is *not* equivalent to 12 : 8?

**A.** 3 : 2 **B.** 6 : 4 **C.** 24 : 16 **D.** 2 : 3

**12. Numerical Response** A $1350 prize is shared in the ratio 2 : 7. The larger share, to the nearest dollar, is $_____.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**13.** Noah and Ella buy a gift for their coach. They pay in the ratio 2 : 5, and Ella pays $21 more than Noah. How much does each pay, and what did the gift cost?

**14.** A punch recipe uses juice and pop in the ratio 3 : 2, and pop and sparkling water in the ratio 4 : 1.

(a) Write the ratio juice : pop : sparkling water. (Hint: make the pop terms match.)

(b) How much juice goes with 1.5 L of sparkling water?

**15.** A painter has a mix of 6 L of white paint and 2 L of blue paint. She wants the ratio of white to blue to be 4 : 1. How much white paint must she add?

## Extra Practice 4: Proportions: Finding the Missing Value

### A. Skills

**1.** Do the two ratios form a proportion? Show how you know.

(a) $5 : 8$ and $15 : 24$ (b) $\frac{6}{9}$ and $\frac{9}{12}$ (c) $2.5 : 1$ and $10 : 4$

**2.** Solve by scaling. Write the $\times$ or $\div$ you used.

(a) $\frac{3}{5} = \frac{x}{40}$ (b) $\frac{7}{x} = \frac{28}{36}$ (c) $x : 12 = 15 : 36$

**3.** Solve by cross-multiplying. Round to the nearest tenth where needed.

(a) $\frac{x}{9} = \frac{5}{6}$ (b) $\frac{14}{x} = \frac{8}{5}$ (c) $\frac{3}{7} = \frac{10}{x}$ (d) $\frac{x}{2.4} = \frac{5}{3}$

**4.** Sasha says the solution of $\frac{5}{x} = \frac{2}{9}$ is $x = 22$. Check her answer with cross products. If it is wrong, find the correct answer.

### B. Applying

**5.** A soup recipe makes 6 servings with 1.5 L of broth. How much broth is needed for 10 servings?

**6.** A model car is built to a scale of $1 : 24$ (1 cm on the model is 24 cm on the real car).

(a) The model is 18.5 cm long. How long is the real car, in metres?

(b) A real pickup truck is 5.4 m long. How long is its model, in centimetres?

**7.** A plant fertilizer label says to mix 5 mL of fertilizer with every 2 L of water. How much fertilizer goes in a 7 L watering can?

**8.** A nurse gives 3 mL of a medicine for every 15 kg of body mass. How much does a 21 kg child get? *(Practice numbers only.)*

**9.** One bag of bark mulch covers 2.5 $m^2$ of garden.

(a) How many bags are needed for a garden of 18 $m^2$?

(b) The bags cost \$6.49 each. What is the cost before tax?

**10.** A photo 15 cm wide and 10 cm tall is enlarged to 36 cm wide without stretching. How tall is the enlargement?

**A.** 54 cm **B.** 24 cm **C.** 31 cm **D.** 2.4 cm

**11. Numerical Response** A dirt bike needs gas and oil mixed in the ratio 32 : 1. The number of millilitres of oil needed for 5 L of gas, to the nearest millilitre, is _____.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**12.** Arjun mixes juice with 2 cans of concentrate for every 7 cans of water. Bea mixes 3 cans of concentrate for every 10 cans of water. Whose juice is stronger? Explain.

**13.** On a map with scale 1 cm : 15 km, a bike route measures 8.4 cm. How long would the same route measure on a different map with scale 1 cm : 40 km?

**14.** A pancake recipe uses 2 cups of flour, 1.5 cups of milk and 2 eggs to make 12 pancakes. Jayden has only 5 eggs (and lots of flour and milk). What is the most pancakes he can make, and how much flour and milk will he need?

## Extra Practice 5: Rates and Unit Rates

### A. Skills

**1.** Decide whether each statement is a rate or a ratio.
(a) 5 red marbles to 7 blue marbles _____ (b) 3 goals in 2 games _____
(c) 500 mL of juice for 4 people _____ (d) 40 cm to 60 cm _____

**2.** Find each unit rate. Include the units.
(a) 325 km in 5 h

(b) $114 for 6 h of work

(c) 264 words in 4 min

(d) 9 L in 2.5 min

**3.** Find each heart rate in beats per minute.
(a) 16 beats in 15 s

(b) 30 beats in 20 s

**4.** Convert each rate.
(a) 7.2 km/h to m/min

(b) 50 m/min to km/h

### B. Applying

**5.** Noor earns $17.75/h at a grocery store.
(a) How much does she earn in a 6-hour shift?

(b) How many hours must she work to earn $710.00?

**6.** The school printer prints 84 pages in 4 min. How long will it take to print 525 pages?

**7.** A hardware store weighs screws instead of counting them. 250 screws weigh 1.5 kg. A customer's bag of the same screws weighs 4.2 kg. How many screws are in the bag?

**8.** Ravi's car used 44.1 L of gas to drive 630 km.

(a) What is its fuel consumption in L/100 km?

(b) He plans a 450 km trip. Gas costs \$1.36/L. How much will the gas cost?

**9.** Emma types 336 words in 6 min. Jake types 405 words in 7.5 min. Who types faster, and by how much?

**10.** A car uses 8 L/100 km. How much gas does it need to drive 250 km?

**A.** 3.2 L **B.** 20 L **C.** 31.25 L **D.** 2000 L

**11. Numerical Response** A drip line waters a garden at 4.5 L/h. The number of hours it takes to deliver 117 L of water is _____.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**12.** A slapshot sends a hockey puck at 144 km/h. How many metres per second is that? (Hint: 1 h = 60 × 60 = 3600 s.)

**13.** The Nguyen family is planning a 1200 km summer road trip. Their car uses 6.5 L/100 km and their truck uses 11.5 L/100 km. Gas costs \$1.45/L. How much more would the gas cost if they took the truck?

**14.** Kai runs 10 km in 52 min. Zoe runs at 11.2 km/h. Who is faster? Show your work.

## Extra Practice 6: Unit Price: Finding the Best Buy

### A. Skills

**1.** Find each unit price. Round to the nearest cent.
(a) 5 pairs of socks for $14.95
(b) a 2 kg bag of flour for $5.38
(c) 24 bottles of water for $4.97
(d) 18 eggs for $6.29

**2.** A 200 g bag of chips costs $2.99. A 280 g bag of the same chips costs $3.99. Find the price per 100 g of each. Which is the better buy?

**3.** Change to the same unit, then decide which is the better buy.
(a) orange juice: 900 mL for $3.29, or 2 L for $6.99
(b) cheddar: $1.85 per 100 g at the deli, or a 400 g block for $7.29

**4.** A jar of pasta sauce costs $3.79. The sign says "2 for $7.00".
(a) What is the price of one jar with the deal?

(b) How much do you save on 6 jars with the deal?

**5.** Bulk candy costs $1.48 per 100 g. Complete the price chart.

| **Amount** | 75 g | 200 g | 450 g | 1 kg |
|---|---|---|---|---|
| **Cost** | | | | |

### B. Applying

**6.** A phone repair shop buys a pack of 20 screen protectors for $36.00. It sells them for $14.99 each.
(a) What is the shop's profit on one screen protector?

(b) What is the profit if the shop sells 15 of them?

**7.** The school drama club is ordering T-shirts online. Site A sells 15 shirts for $112.50 plus $15.00 shipping. Site B sells 25 shirts for $195.00 with free shipping. Which site has the lower price per shirt, and by how much?

**8.** A hockey team uses 3 rolls of tape at each practice and has 4 practices a week. A box of 12 rolls costs $29.88.

(a) How many rolls does the team use in a week?

(b) What is the price of one roll?

(c) How much should the team budget for tape for a 10-week season?

**9.** Tile for a kitchen backsplash costs $45.50 per $m^2$. The backsplash is 2.4 m long and 0.5 m high. How much will the tile cost?

**10.** Leah wants to paint one small bathroom door. She needs about 0.8 L of paint. A 3.78 L can costs $54.99 and a 946 mL can costs $19.99.

(a) Find the price per litre of each can.

(b) Which can should Leah buy? Give a reason that is not the unit price.

**11.** Which jug of laundry detergent is the best buy?

**A.** 1.47 L for $7.97 **B.** 2.95 L for $14.97 **C.** 4.43 L for $22.97 **D.** 5.9 L for $31.47

**12. Numerical Response** A 9 kg bag of cat litter costs $13.95. At the same unit price, the cost of 4 kg of litter, to the nearest cent, is $_____.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**13.** A 650 mL bottle of shampoo costs $7.80. What price would a 1 L bottle need to have the same unit price? What does that tell you about a 1 L bottle priced at $11.49?

**14.** Laundry pods come in a 24-pack for $8.99 or a 42-pack for $14.99. Tess has a coupon for $2.00 off the 24-pack.

(a) Without the coupon, which pack is the better buy?

(b) With the coupon, which pack is the better buy?

**15.** Juice comes in 1 L cartons for \$2.89 and 2 L bottles for \$5.49. Omar needs exactly 3 L for a party.

(a) Find the cost of three 1 L cartons, and of one 2 L bottle plus one 1 L carton. Which is cheaper?

(b) Two 2 L bottles give 4 L. How much more is that than the cheaper 3 L choice? Why might Omar still choose it?

## Extra Practice 7: Percents Refresher: Percents, Decimals and Percent of an Amount

### A. Skills

**1.** Write each percent as a decimal.

(a) 56% _____ (b) 3% _____ (c) 4.5% _____

(d) 110% _____ (e) 0.2% _____ (f) 400% _____

**2.** Write each decimal or fraction as a percent.

(a) 0.71 _____ (b) 0.08 _____ (c) 2.5 _____

(d) 0.015 _____ (e) $\frac{9}{10}$ _____ (f) $\frac{3}{8}$ _____

**3.** Without a calculator, find each percent of $36.00.

(a) 10% _____ (b) 5% _____ (c) 15% _____

(d) 1% _____ (e) 25% _____ (f) 30% _____

**4.** Calculate.

(a) 25% of 340 (b) 9% of 600 (c) 120% of 45 (d) 0.8% of 750

**5.** Find each percent. Round to the nearest tenth of a percent if needed.

(a) 12 of 48 (b) 33 of 60 (c) 70 of 56 (d) 4 of 7

### B. Applying

**6.** Ella's phone plan has 8 GB of data each month. She has used 85% of it.

(a) How many gigabytes has she used?

(b) How many gigabytes are left?

**7.** A goalie stopped 34 of 37 shots in one game and 28 of 30 shots in the next game. Find the save percent for each game, to the nearest tenth. In which game did the goalie do better?

**8.** Desmond earns \$16.40 per hour at a hardware store. He gets a 3.5% raise.
(a) How much more will he earn per hour?

(b) What is his new hourly wage?

**9.** A 30 kg bag of concrete mix is 15% cement by mass. How many kilograms of cement are in the bag? How many kilograms are not cement?

**10.** Owen's lunch bill is \$38.60. He can't decide between a 15% tip and a 20% tip.
(a) Find both tips.

(b) How much more is the 20% tip? What is the total with the 20% tip?

**11.** Which of these is *not* equal to 40%?
**A.** 0.4 **B.** $\frac{2}{5}$ **C.** 0.04 **D.** $\frac{40}{100}$

**12. Numerical Response** A school sold 225 of its 360 tickets for a concert. The percent of the tickets sold, to the nearest tenth, is _____%.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**13.** Lena left a 15% tip of \$6.30. What was her bill before the tip? (Hint: \$6.30 is 15% of the bill. What is 5%? What is 100%?)

**14.** (a) Find 30% of 50 and 50% of 30. What do you notice?
(b) Use what you noticed to find 4% of 75 in your head.

**15.** Kai's phone is fully charged (100%). Watching video uses 12% of the battery each hour. After how many full hours of video will the battery first be below 20%? Show the battery level each hour.

## Extra Practice 8: Setting a Price: Markup and Sales Tax

### A. Skills

**1.** Find the markup and the retail price.
(a) cost \$15.00, markup 60% (b) cost \$250.00, markup 35% (c) cost \$4.75, markup 200%

**2.** Find the markup percent.
(a) cost \$80.00, retail \$100.00 (b) cost \$2.50, retail \$6.00

**3.** Find the GST and the total cost in Alberta.
(a) a \$36.00 video game gift card (b) a \$149.99 pair of running shoes

**4.** Write the number you multiply by to get the new price.
(a) a 45% markup _____ (b) Alberta GST _____
(c) Ontario HST (13%) _____ (d) BC GST + PST (12%) _____

### B. Applying

**5.** An auto parts store pays \$9.80 for a wiper blade and marks it up 75%. What is the retail price?

**6.** A food truck raises the price of a poutine from \$11.50 to \$12.75. What is the percent increase, to the nearest tenth?

**7.** A lawn mower costs \$379.00 in Regina, Saskatchewan (5% GST and 6% PST). Find the GST, the PST and the total.

**8.** An electrician in Lethbridge buys 3 light fixtures at \$47.00 each. She charges her customer 25% more than she paid, plus GST. What does the customer pay for the fixtures?

**9.** A laptop costs \$899.99 in Halifax (14% HST) and \$949.99 in Edmonton (5% GST). Where is it cheaper after tax, and by how much?

**10.** A store pays \$60.00 for a lamp and sells it for \$90.00. The markup percent is

**A.** 33.3% **B.** 50% **C.** 30% **D.** 150%

**11. Numerical Response** The PST in British Columbia (7%) on a \$52.30 purchase, to the nearest cent, is \$_____.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**12.** A receipt from a Calgary store shows a total of \$63.00, with GST included. What was the price before tax? (Hint: total = price $\times$ 1.05, so undo the multiplying.)

**13.** A factory sells a toaster to a distributor for \$20.00. The distributor marks it up 25% and sells it to a store. The store marks that price up 40%.

(a) What is the store's retail price?

(b) What is the total markup percent from the factory price? Is it $25\% + 40\% = 65\%$?

**14.** A bike sells for \$420.00 before tax. The store's markup was 75%. What did the store pay for the bike? (Hint: retail = cost $\times$ 1.75.)

## Extra Practice 9: On Sale! Discounts and Sale Prices

### A. Skills

**1.** Find the discount and the sale price.
(a) \$36.00, 25% off (b) \$129.50, 10% off (c) \$8.48, 50% off

**2.** What decimal do you multiply by to find the sale price?
(a) 45% off _____ (b) 8% off _____ (c) 70% off _____ (d) 33% off _____

**3.** Find the percent decrease.
(a) \$20.00 to \$16.00 (b) \$64.00 to \$56.00 (c) \$3.60 to \$2.70

**4.** In Lethbridge, a lamp is regularly \$42.00 and is 20% off. Find the total cost with 5% GST.

### B. Applying

**5.** A mobile mechanic charges \$95.00 per hour and gives seniors 10% off. What does a senior in Airdrie pay for a 3-hour job, including 5% GST?

**6.** A phone charger costs \$34.99 at Store A with 40% off, and \$26.99 at Store B with \$5 off. Which is cheaper, and by how much?

**7.** A box of cereal is \$5.79. One store has "Buy one, get one 50% off". Another has "3 for the price of 2".
(a) Find the percent saved with each deal.

(b) Give one reason the deal with the smaller percent could still be the better choice.

**8.** On a trip to Halifax, Nova Scotia (14% HST), Daniel buys winter boots regularly $180.00 at 30% off. What is his total cost, and how much does he save compared with the regular price plus HST?

**9.** A laptop is regularly $900.00. It is 15% off, and students get an extra 10% off the sale price. Find the final price and the total percent off.

**10.** A hoodie was $60.00. A store raised the price to $75.00 and then put up a "20% off" sign. Is the sale a real deal? Explain.

**11.** A $250.00 bike is on sale for $200.00. The percent decrease is

**A.** 20% **B.** 25% **C.** 50% **D.** 80%

**12. Numerical Response** A store buys a jacket wholesale for $30.00, marks it up 60%, then later sells it at 25% off. The sale price, to the nearest dollar, is $_____.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**13.** After 30% off, a jacket costs $63.00. What was the regular price? (Hint: the sale price is 70% of the regular price.)

**14.** Which is the bigger discount on a $120.00 item: 25% off then an extra 20% off, or a single 40% off?

**15.** Kai says, "It's cheaper if the store takes 30% off first and then adds GST." Lena says, "It's cheaper if the store adds GST first and then takes 30% off." Test both on an $80.00 item. Who is right?

## Extra Practice 10: Currency Exchange: Converting, and Buy vs Sell Rates

### A. Skills

**1.** The rate is 1 CAD = 0.725 USD. Change each amount into US dollars. (a) $60.00 CAD (b) $384.00 CAD

**2.** The rate is 1 USD = 1.38 CAD. Find each price in CAD. (a) A coffee, $4.50 USD (b) A tablet, $575.00 USD

**3.** The only rate posted is 1 EUR = 1.55 CAD.
(a) How many euros do you get for $240.00 CAD?

(b) A museum ticket costs 19.80 EUR. What is that in CAD?

**4.** Use these rates: 1 CAD = 108 JPY, 1 JPY = 0.0093 CAD, 1 CAD = 13.3 MXN, 1 MXN = 0.075 CAD.
(a) $45.00 CAD into yen (b) 12 000 JPY into CAD

(c) 650 MXN into CAD (d) $95.00 CAD into pesos

**5.** The rate is 1 CAD = 0.549 GBP.
(a) Is one British pound worth more or less than one Canadian dollar? How do you know?

(b) How many pounds do you get for $50.00 CAD?

### B. Applying

**6.** The rate is 1 USD = 1.38 CAD. Estimate each price in CAD in your head, then find the exact price. (a) A phone case, $27.00 USD (b) Two nights in a motel, $210.00 USD

**7.** Use this bank board (CAD for 1 unit).

| Currency | We buy | We sell |
|---|---|---|
| USD | 1.35 | 1.41 |
| GBP | 1.77 | 1.87 |

(a) Rosa is going to Scotland. How much does 180 GBP in cash cost her in CAD?

(b) She comes home with 45 GBP and sells them back. How much CAD does she get, and how much did she lose on those 45 pounds?

**8.** Use the board in Question 7. Liam needs \$250.00 USD. The bank charges no fee. A currency kiosk uses 1 USD = 1.38 CAD but charges a 4% commission. Which is cheaper, and by how much?

**9.** Aiden finds earbuds on a US website for \$24.99 USD plus \$8.00 USD shipping. His card uses 1 USD = 1.38 CAD and adds a 2.5% fee. A Calgary store sells the same earbuds for \$39.99 plus 5% GST. Which is cheaper, and by how much? (Ignore any charges at the border.)

**10.** A blanket at a market in Oaxaca costs 520 MXN. The only rate posted is 1 CAD = 13.3 MXN. The cost in CAD is
**A.** \$6916.00 **B.** \$39.10 **C.** \$52.00 **D.** \$3.91

**11. Numerical Response** A bank sells euros at 1 EUR = 1.60 CAD. Jun changes \$850.00 CAD into euros. To the nearest euro, he gets _____ EUR.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**12.** Tess is in Texas with \$200.00 USD. She is flying on to Mexico. Use 1 USD = 1.38 CAD and 1 CAD = 13.3 MXN to find how many pesos her US dollars are worth. (Hint: change to CAD first.)

**13.** A jacket costs \$129.99 plus 5% GST in Calgary. Its US online price includes free shipping. Aisha's card uses 1 USD = 1.38 CAD and adds a 2.5% fee. What is the highest US price (round down to the cent) that would still cost her no more than the Calgary store?

**14.** A bank buys US dollars at 1.35 and sells them at 1.41 (CAD for 1 USD). You change \$1000.00 CAD into US dollars and change it all back the same day. What percent of your money do you lose? Round to the nearest tenth of a

percent.

## Extra Practice 11: Unit 1 Review

**Practice Test 2.** Treat this like the unit test. Show your work and round money to the nearest cent. Rates in this booklet are examples.

### A. Skills

**1.** Write each ratio in simplest form.
(a) 16 : 40 (b) 250 mL : 1 L (c) 2.4 : 0.8

**2.** Solve each proportion.
(a) $\frac{x}{8} = \frac{15}{24}$ (b) $\frac{6}{x} = \frac{4}{14}$ (c) $2.5 : 4 = x : 10$

**3.** Find each unit rate.
(a) \$87.00 for 6 hours of babysitting (\$/h) (b) 455 km in 5 h (km/h) (c) 36 L of gas for 480 km (L/100 km)

**4.** (a) Find the 5% GST on \$86.40. (b) A \$54.00 sweater is 25% off. Find the discount and the sale price. (c) Find 12.5% of \$240.00.

**5.** Use 1 USD = 1.38 CAD.
(a) Change \$75.00 USD to Canadian dollars. (b) Change \$207.00 CAD to US dollars.

### B. Applying

**6.** Three roommates share a \$1950.00 monthly rent in the ratio 5 : 4 : 6, based on the size of their bedrooms. How much does each person pay?

**7.** A 10 cm by 15 cm photo is enlarged so that the short side is 24 cm. How long is the long side?

**8.** Shampoo comes in three sizes: 400 mL for \$5.49, 750 mL for \$8.99 and 1 L for \$13.49.

(a) Find the price per 100 mL of each size. Which is the best buy?

(b) Give one reason that is **not** about price for buying the 400 mL bottle.

**9.** A Calgary store pays \$1020.00 for a shipment of 12 jackets.

(a) What is the wholesale cost of one jacket?

(b) The store marks each jacket up 65%. What is the retail price?

(c) What does a customer pay with 5% GST?

(d) What would the same jacket cost in Ontario (13% HST)?

**10.** Kenji buys 4 winter tires at \$189.00 each. The shop gives 15% off, and then he uses a coupon for \$40.00 off. GST is added last. What is his total?

**11.** A bank shows these rates, in Canadian dollars for 1 unit of each currency.

| **Currency** | **We buy** | **We sell** |
|---|---|---|
| US dollar (USD) | 1.34 | 1.42 |
| Mexican peso (MXN) | 0.072 | 0.081 |

(a) Dani buys 4000 Mexican pesos before a trip to Puerto Vallarta. How much does she pay in Canadian dollars?

(b) She brings home 650 pesos and sells them back. How much does she get?

(c) How much money did she lose on those 650 pesos by changing them there and back?

**12.** The price of a movie ticket goes from \$25.00 to \$30.00. The percent increase is

**A.** 5% **B.** 16.7% **C.** 20% **D.** 83.3%

**13.** In Japan, 1 CAD = 108 JPY. A meal costs 5400 yen. In Canadian dollars, that is

**A.** \$50.00 **B.** \$583 200.00 **C.** \$5.00 **D.** \$54.00

**14.** Jeans are $40.00 a pair. Which promotion gives the lowest price for 2 pairs?

**A.** Buy one, get the second half price

**B.** 25% off each pair

**C.** $25 off when you buy 2 pairs

**D.** A and B tie for the lowest price

**15. Numerical Response** A car uses 6.4 L/100 km. The fuel needed for a 285 km trip, in litres to the nearest tenth, is _____.

| | | | |
|---|---|---|---|
| | | | |

**16. Numerical Response** A store pays $16.00 for a phone charger and sells it for $26.00. The markup percent, to the nearest tenth, is _____%.

| | | | |
|---|---|---|---|
| | | | |

**C. Stretch**

**17.** In Seattle, a jacket costs $84.00 USD plus 10% sales tax. In Calgary, the same jacket costs $119.99 plus 5% GST. Use 1 USD = 1.38 CAD.

(a) Estimate the Seattle cost in Canadian dollars in your head.

(b) Find both costs exactly in Canadian dollars. Which is cheaper, and by how much?

(c) The prices are very close. Name one other thing that could help you decide.

**18.** After 30% off, a backpack's sale price is $63.00. What was the regular price? (Hint: what percent of the regular price is $63.00?)

**19.** Concrete is mixed from cement, sand and gravel in the ratio $1 : 2 : 3$. A crew needs 1.8 $m^3$ of mix for a shed pad.

(a) How much of each material do they need?

(b) A bag of cement holds 0.03 $m^3$. How many bags do they need?